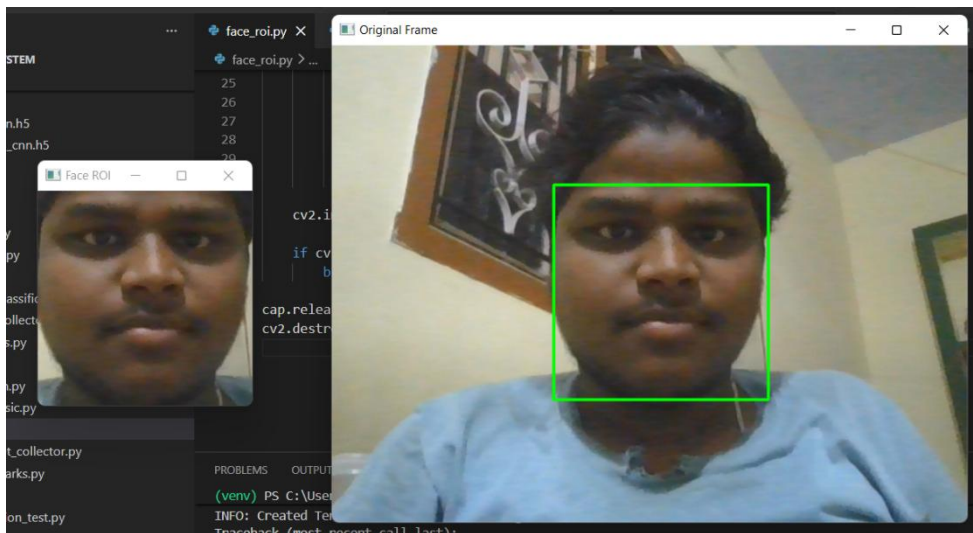


APPENDIX - II

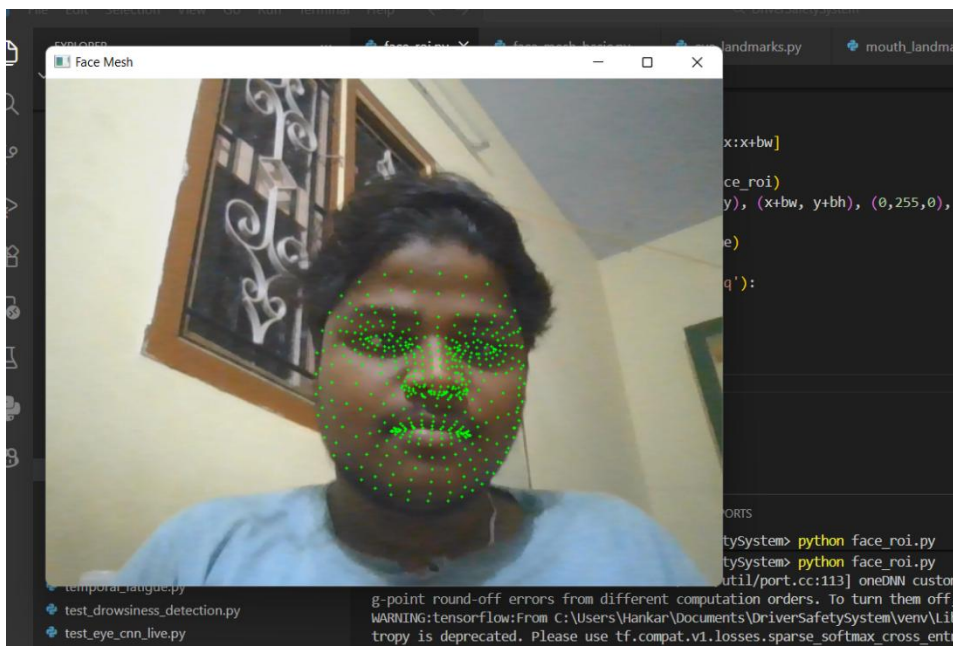
SCREENSHOTS

1. Face Detection Output

This screenshot shows the detection of the driver's face within the video frame. A bounding box is drawn around the detected face, indicating that the system has successfully identified the region of interest. This step ensures that only the facial region is processed for further analysis.



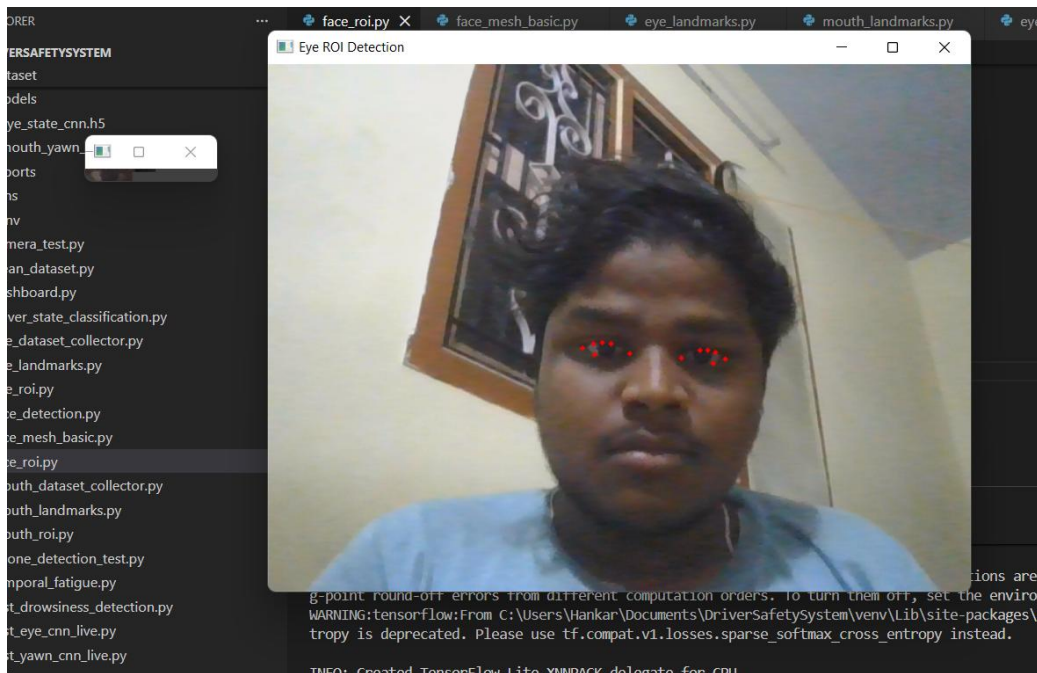
2. Eye Region Extraction



This screenshot shows the cropped eye region obtained from the detected face. The system isolates the eye area to analyze eye state (open or closed). Proper extraction of this region is important for accurate classification by the deep learning model.

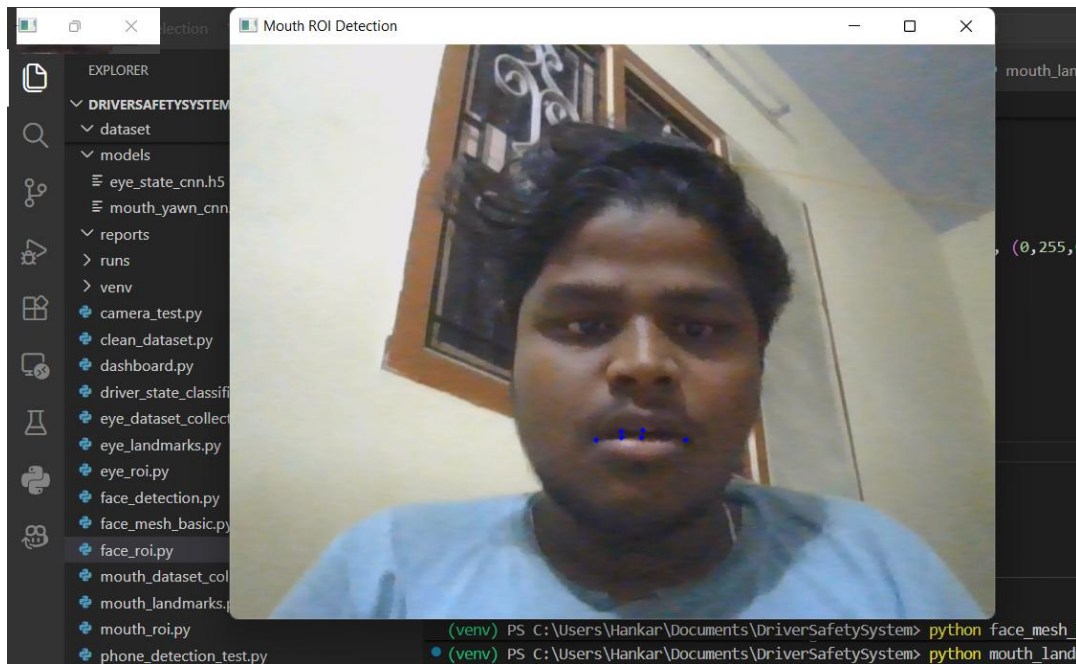
3. Eye Region Extraction

This screenshot shows the cropped eye region obtained from the detected face. The system isolates the eye area to analyze eye state (open or closed). Proper extraction of this region is important for accurate classification by the deep learning model.



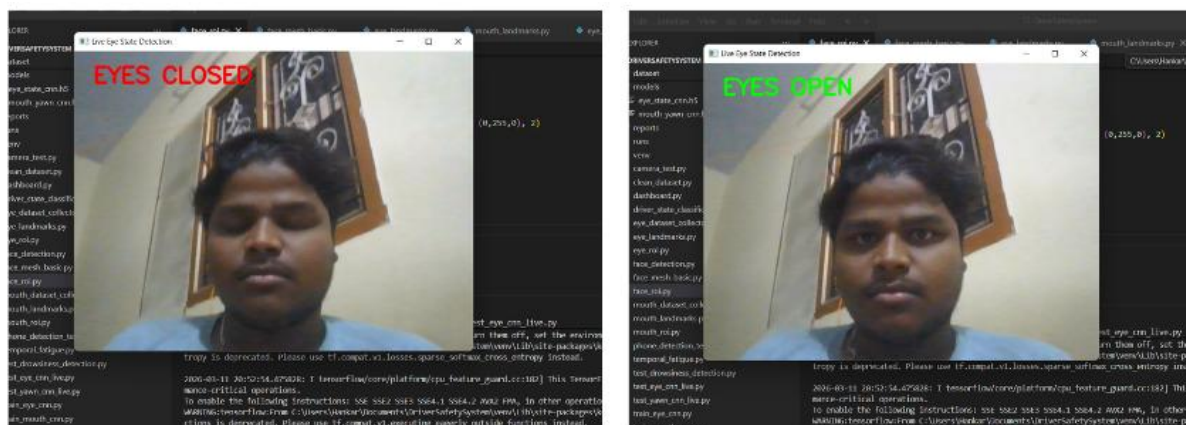
4. Mouth Region Extraction

This screenshot displays the extracted mouth region used for yawning detection. The system identifies and isolates the mouth area to determine whether the driver is yawning. This helps in improving fatigue detection accuracy.



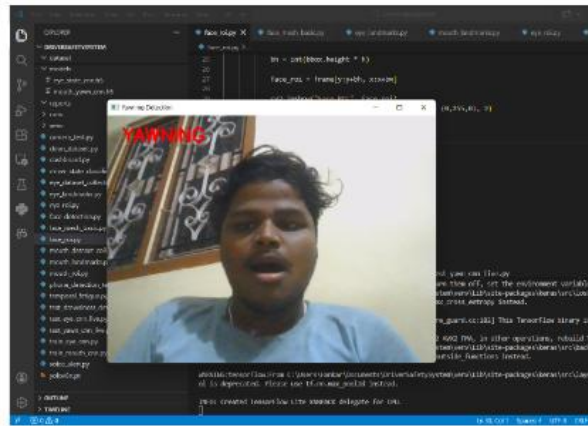
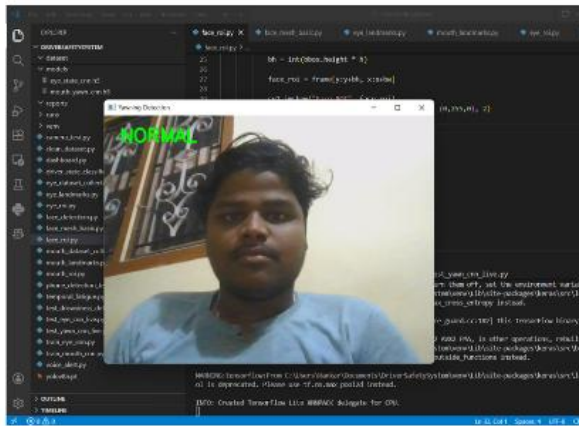
5. Eye State Detection Output

This screenshot shows the classification result of the eye detection model. The system identifies whether the driver's eyes are open or closed and displays the result on the screen. Accurate prediction confirms the effectiveness of the trained CNN model.



6. Yawning Detection Output

This screenshot presents the output of the yawning detection model. The system detects whether the driver is yawning and displays the corresponding result. This feature helps in identifying signs of fatigue.



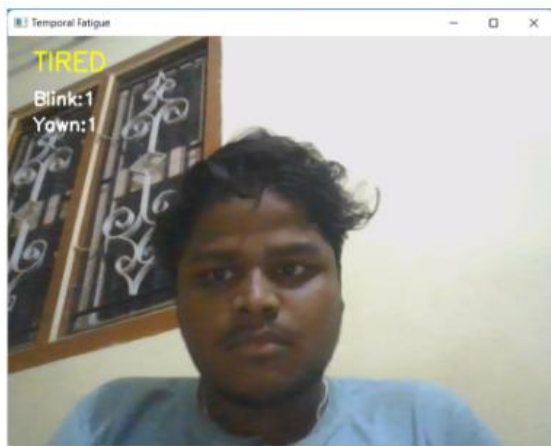
7. Mobile Phone Detection

This screenshot presents the output of the mobile phone detection model. The system detects whether the driver is using mobile phone and displays the corresponding result. This feature helps in identifying signs of distraction.



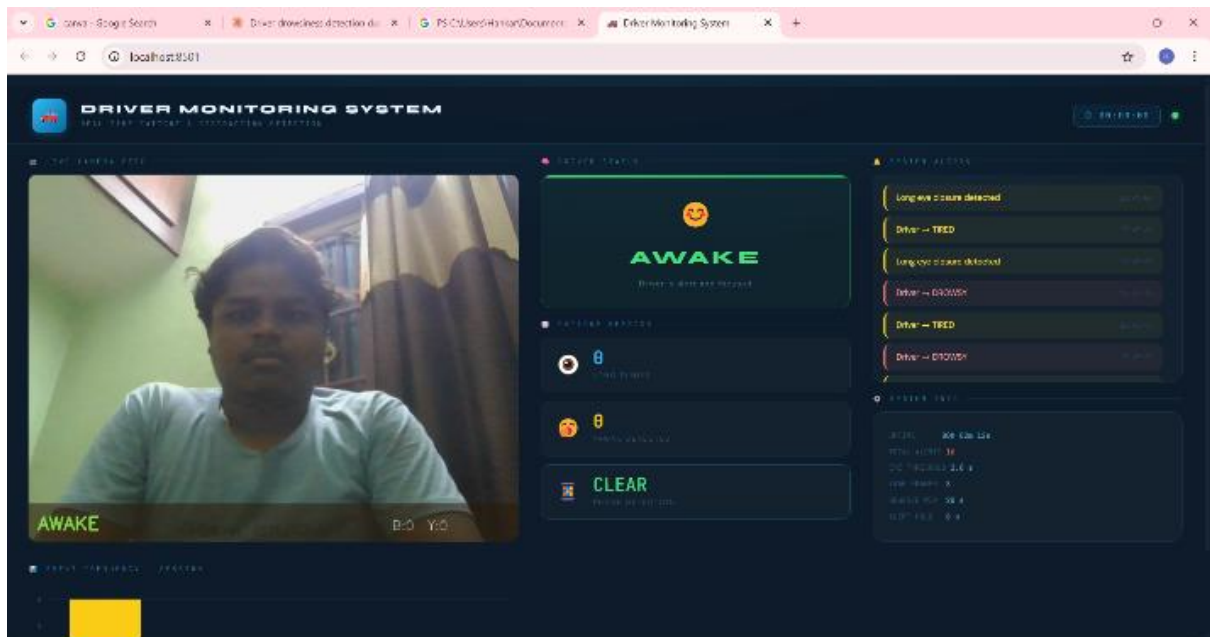
8. Drowsiness Detection Output

This screenshot shows the final driver state classification based on combined indicators. The system evaluates eye closure and yawning patterns to determine whether the driver is in a drowsy state. The result is displayed clearly for user understanding.



9. Dashboard Visualization

This screenshot shows the dashboard interface displaying real-time driver status. It provides a clear view of the system output, including driver state and alerts. The dashboard enhances usability and monitoring.



10. System Performance Output

This screenshot presents the model performance results such as accuracy or confusion matrix. It demonstrates the effectiveness of the trained models in detecting drowsiness with high accuracy.

```

===== Eye State Detection Model =====
WARNING:tensorflow:From C:\Users\Hankar\Documents\DriverSafetySystem\venv\Lib\site-packages\keras\src\utils\tf_utils.py:110: tf.nn.conv2d is deprecated. Please use tf.compat.v1.nn.conv2d instead.

48/48 [=====] - 26s 407ms/step - loss: 0.0170 - accuracy: 0.9961
Eye Model Accuracy: 99.6073305606842 %
48/48 [=====] - 3s 56ms/step

Confusion Matrix (Eye Model)
[[746  0]
 [ 6 776]]

Classification Report (Eye Model)

```

	precision	recall	f1-score	support
0	0.99	1.00	1.00	746
1	1.00	0.99	1.00	782
accuracy			1.00	1528
macro avg	1.00	1.00	1.00	1528
weighted avg	1.00	1.00	1.00	1528

```

===== Yawning Detection Model =====
19/19 [=====] - 8s 413ms/step - loss: 0.0098 - accuracy: 0.9934
Mouth Model Accuracy: 99.33775067329407 %
19/19 [=====] - 1s 42ms/step

Confusion Matrix (Mouth Model)
[[300  0]
 [ 4 300]]

Classification Report (Mouth Model)

```

	precision	recall	f1-score	support
0	0.99	1.00	0.99	300
1	1.00	0.99	0.99	304
accuracy			0.99	604
macro avg	0.99	0.99	0.99	604
weighted avg	0.99	0.99	0.99	604